Dr Diane Vassallo

B.Ed.(Hons)(Melit.) · M.Sc.(Lancs.) · Ph.D.(Belfast)
Lecturer & Researcher | Department of Technology and Entrepreneurship Education
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PROFILE

Full-Time Resident Academic Lecturer at the University of Malta's Faculty of Education, specialising in Computing and Computational Thinking Education. Research spans CT integration in early childhood, curriculum design for digital literacy, executive functions, and AI in Education. Co-founder of Computational Thinking Malta and co-organiser of the national Bebras Challenge. Lead and partner on multiple Erasmus+ funded projects totalling over €2.25 million. Committed to preparing future educators for the challenges of teaching in a digital age.

RESEARCH INTERESTS

- Computational Thinking and its cognitive foundations, including links to Executive Functions
- Artificial Intelligence in Education: ethics, creativity, and teacher/learner agency
- Curriculum studies with a focus on ICT, entrepreneurship, and policy analysis in small states
- Innovative pedagogies for digital literacy, computational thinking, and sustainability education

CURRENT POSITION

Jun 2021 – present

Full-Time Resident Academic Lecturer

University of Malta, Faculty of Education — Department of Technology and Entrepreneurship Education

- Design and deliver modules in CT integration, digital literacy curriculum design, and programming pedagogy
- Supervise Masters of Teaching and Learning dissertations (currently supervising 4 students)
- Act as examiner on Masters dissertations across the Faculty of Education
- Serve on faculty and university committees
- Teaching Practice Examiner — supervise and assess student teachers on school placement

Feb 2026 – present

Voluntary International Teaching Engagement

Online (Pakistan)

Deliver online teaching in Computer Science and Computing Pedagogy to in-service teachers in Pakistan.

Sep 2025 – Jan 2026

Community Outreach — National Council of Women

Online (Malta)

Co-ordinated and taught an entrepreneurship course for the National Council of Women, supporting women's professional development and digital literacy.

RESEARCH FUNDS & EU PROJECTS

Dec 2025 – Nov 2027

ALPER-EU | Erasmus+ KA220-SCH-B5FEF9B5 | €400,000

K12 Agile Learning of Programming with Educational Robotics

A European cooperation project bringing together policy makers, universities, and primary schools across five EU countries to strengthen early programming and robotics education through an agile

learning (scrum-style) approach. Develops a standardised, competence-based curriculum built around 60 gamified robotics sprints, an engagement assessment tool, and an asynchronous teacher-training platform. Piloted in real classrooms including learners with ADHD, with EU-wide dissemination via the SCIENTIX network (51 countries).

Apr 2025 – May 2028 **EmpowerAId** | ERASMUS-EDU-2024-PEX-TEACH-ACA | €1,200,000

Empowering Pre- and In-Service Teachers to Harness the Potential of AI for Designing Creative and Ethical Digital Education Ecosystems

Builds the AI literacy, ethical awareness, and digital competence of educators through co-created, inclusive, and future-ready training programmes. Promotes the integration of AI in education through participatory design, piloting, and evaluation of learning pathways tailored to teachers' real-world needs.

Dec 2024 – May 2027 **GreenToC** | Erasmus+ 2024-1-RO01-KA220-HED-000252839 | €400,000

Synergies Between Higher Education Institutions and the Labour Market for Developing Sustainability Capabilities for Green Transformation Strategies Using the Theory of Change

Bridges higher education and the labour market using the Theory of Change framework to co-create and embed sustainability capabilities into university curricula and organisational strategies.

Dec 2024 – present **SEA-EU Task 2.2** | European University Alliance

Contributes to the SEA-EU Task 2.2 initiative by supporting the design and delivery of interdisciplinary learning connecting digital innovation with green transformation. Focuses on practical resources, staff development, and evidence-informed approaches across partner institutions.

Dec 2022 – May 2025 **CThink.IT** | Erasmus+ 2022-1-MT01-KA220-SCH-000086903 | €250,000 | Completed

Think, Learn and Play in a Computational Thinking Way

Led by the University of Malta in a consortium of 6 European partners. Investigated the state of the art of CT in early years education and developed an unplugged board game to teach computational thinking skills to young children.

TEACHING PORTFOLIO

Current Study Units — University of Malta

- TEM5016 — Introduction to Computational Thinking
- TEM5007 — Resources for Teaching Computing
- TEM5013 — An Introduction to the Practice of Computing Education, Part 1
- TEM5015 — An Introduction to the Practice of Computing Education, Part 2
- TEM5014 — The Practice of Assessment in Computing Education
- TEM5018 — Robots, Programmable Toys and Physical Computing
- LLI5025 — Digital Pedagogies: A Hands-On Approach
- ICT5101 — Internet of Things
- LLI5001 / LLI5002 — Field Placement 1 & 2

Previous Teaching — Institute for Education (Sep 2020 – Sep 2021)

- Developed and delivered pedagogy modules for the Master of Education in Computing, ICT and Vocational and Applied IT, accredited by NCFHE
- Modules spanning pedagogy of Programming, Creative Design, and Technical Training in Computing Education

PUBLICATIONS

2026

Vassallo, D. (2026). The AI guilt complex: Moral emotions and ethical dilemmas in academic technology adoption. *Journal of Academic Ethics.* Advance online publication. https://doi.org/10.1007/s10805-026-09726-3

Juuti, K., Turkkila, M., Usenik, J., Vassallo, D., & Stupurienė, G. (2026, March). Teacher educators' emotions toward generative AI: Cross-national focus groups. *The Nordic Educational Research Association (NERA) Conference*, VIA University College, Aarhus, Denmark, 4–6 March 2026, pp. 1–7. https://www.um.edu.mt/library/oar/handle/123456789/144776

In Press

Vassallo, D. & Frendo, C.S. (in press). Cognitive abilities and computational thinking in intellectually high-ability adolescents: A mixed-methods study. *Australasian Journal of Gifted Education.*

2025

Vassallo, D., Govus, G., & Busuttil, L. (2025). Between ambition and implementation: A comparative study of Malta's ICT C3 curriculum in the context of small-state digital education.

Busuttil, L., & Vassallo, D. (2025). Evaluating the ICT C3 curriculum in Maltese schools: Teacher perspectives. *University of Malta, Faculty of Education.*

Busuttil, L., Vassallo, D., & Schembri, P. (2025). Computational thinking: Exploring approaches in early childhood education. In *Effective Computer Science Education in K-12 Classrooms* (pp. 27–54).

Frendo, C. S., & Vassallo, D. (2025). Educational neuroscience meets AI: A framework for secondary science teaching. *Malta Journal of Education, 6*(1), 5–21.

Vassallo, D. (2025). Fostering computational thinking in early learners: An iterative approach in a Maltese primary school. *Discover Education, 4*(1), 1–19.

Arslankara, V. B., Owusu Ansah, L., Vassallo, D., Thériault, C., & Pivetta da Silva, M. (2025). Yapay zeka yeterliği, bilişimsel düşünme ve sanal risk algısı arasındaki ilişkide dijital iyi oluşun aracı ve düzenleyici rolü [The mediating and moderating role of digital well-being in the relationship between AI competence, computational thinking, and virtual risk perception]. *5th International Conference on Educational Technology and Online Learning – ICETOL 2025*, Cunda, Ayvalık, Turkey, 26–29 August 2025, p. 310. ISBN 9789756993378. https://www.um.edu.mt/library/oar/handle/123456789/144559

2023

Bratitsis, T., Tsapara, M., Melliou, K., Busuttil, L., Vassallo, D., Callus, J., et al. (2023). Cultivating computational thinking in early years through board games: The Cthink.it approach. In *Interactive Mobile Communication, Technologies and Learning* (pp. 78–89).

Busuttil, L., Vassallo, D., Callus, J., Bratitsis, T., Tsapara, M., Melliou, K., et al. (2023). Promoting computational thinking in early childhood: A novel board game design. *ICERI2023 Proceedings*, 4868–4876.

Vassallo, D. (2023). Embarking on computational thinking. *ICERI2023 Proceedings*, 5083–5088.

Vassallo, D., & Busuttil, L. (2023). Integrating computational thinking into classroom practice: A case study. In *To Be or Not to Be a Great Educator* (p. 600).

Bratitsis, T., Busuttil, L., Vassallo, D., Koliakou, I., Tsapara, M., & Melliou, K. (2023). Cthink.it: A board-game for computational thinking in early years. *Academic Conferences International Limited.*

View full publication list on UM Open Access Repository →

THESES & DISSERTATIONS

Foundational academic works underpinning my research trajectory.

2013

Doctor of Philosophy — Doctoral Thesis

Queen's University Belfast, Belfast, United Kingdom

Buhagiar, D. (2013). *Mainstream technologically-enhanced learning environments in special schools in Northern Ireland.* (Doctoral dissertation, Queen's University Belfast.)

Published under previous name.

2004

MSc Dissertation

Lancaster University, Lancaster, United Kingdom

Vassallo, D. (2004). *Designing online instruction: Incorporating the elements of meaningful learning in the design of an e-learning course.* (MSc dissertation, Lancaster University.) MSc in Advanced Learning Technology.

2001

B.Ed. (Hons.) Dissertation

University of Malta, Msida, Malta

Cefai, D., & Vassallo, D. (2001). *English language verbs: Software and teaching pack.* (B.Ed. Honours dissertation, University of Malta.) A multimedia software package designed to support secondary school students in Forms I and II in learning English verb concepts; evaluated with 40 students across four schools.

CONFERENCE PRESENTATIONS

Mar 2026

NERA 2026 — Nordic Educational Research Association Conference — VIA University College, Aarhus, Denmark

Teacher educators' emotions toward generative AI: Cross-national focus groups

With Juuti, K., Turkkila, M., Usenik, J., & Stupurienė, G. (EmpowerAId project) · Peer-reviewed

Aug 2025

ICETOL 2025 — 5th International Conference on Educational Technology and Online Learning — Cunda, Ayvalık, Turkey

Yapay zeka yeterliği, bilişimsel düşünme ve sanal risk algısı arasındaki ilişkide dijital iyi oluşun aracı ve düzenleyici rolü

(The mediating and moderating role of digital well-being in the relationship between AI competence, computational thinking, and virtual risk perception)

With Arslankara, V.B., Owusu Ansah, L., Thériault, C., & Pivetta da Silva, M. · Cross-national study: Turkey, Malta, Ghana, Canada, Brazil · Peer-reviewed

Nov 2023

ICERI 2023 — Seville, Spain

Embarking on Computational Thinking (Web of Science core collection)

Oct 2022

ATEE 2022 — Morschach, Switzerland

Integrating Computational Thinking into Classroom Practice: A Case Study (Web of Science core collection)

Sep 2011

BERA Annual Conference — London, UK

ICT integration and Northern Ireland's special education: A survey study; Giving students with learning difficulties a 'voice'

Sep 2009

ECER 2009 — Vienna, Austria

Using Activity Theory to reveal the barriers to technology-enhanced learning in special school classrooms; ICT integration and Northern Ireland's special education: A survey study

Sep 2009 **BERA Annual Conference** — Manchester, UK

Exploring the effects of a technologically-enhanced learning environment in a classroom for students with learning difficulties: An activity theory approach

Sep 2008 **ECER 2008** — Göteborg, Sweden

Digital portfolios and online voting systems: Best practices in assessment for learning and the promotion of self-monitoring skills

Sep 2008 **BERA Annual Conference** — Edinburgh, UK

Digital portfolios and online voting systems: Best practices in assessment for learning and the promotion of self-monitoring skills

Jun 2008 **ED-MEDIA 2008** — Vienna, Austria

Digital portfolios and online voting systems: Best practices in assessment for learning and the promotion of self-monitoring skills

POLICY ENGAGEMENT & PUBLIC CONTRIBUTION

2024 **Invited Panelist — AI in Education Public Event, Malta**

Contributed expert perspectives on computational thinking, AI literacy, and ethical technology integration in schools at a public forum on AI in Education.

2024 **Public Consultation Panel — Digital Education Strategy 2030**

DDLTS (Directorate for Digital Literacy and Transversal Skills), Government of Malta

Invited contributor and panel participant at the public consultation on Malta's Digital Education Strategy 2030. Contributed expert perspectives on digital literacy, computational thinking, and ICT curriculum policy.

2024 **Working Panel — Digital Education Strategy 2030**

DDLTS, Ministry for Education, Government of Malta

Participated in working group discussions for the development of Malta's Digital Education Strategy 2030.

ACADEMIC SERVICE & COMMITTEE MEMBERSHIP

National Curriculum & Syllabus Panels — MATSEC, University of Malta

2024 – 2026 **AM/IM Computing Syllabus Panel** — MATSEC, University of Malta

Member of the panel responsible for developing and reviewing the Advanced and Intermediate Level Computing syllabi (AM07/IM07), launched 2025. Contributed to syllabus content review, query resolution, and approval processes through the MATSEC Board and Senate.

2016 – 2019 **AM/IM Computing Syllabus Panel** — MATSEC, University of Malta

Member of the panel for the previous iteration of the Advanced and Intermediate Level Computing syllabus.

2021 – 2022 **SEC Computing Syllabus Panel** — MATSEC, University of Malta

Member of the panel responsible for reviewing and updating the SEC (O Level) Computing syllabus. Changes proposed by the panel were approved by the MATSEC Board and Senate in March 2022, with the updated SEC LOF Computing Syllabus published for 2025.

University Committees — Faculty of Education, University of Malta

2025 – 2026	**Board of Studies** — Postgraduate Certificate in Computational Thinking and Coding for Primary Educators **Students' Request Sub-Committee** **Field Placement Committee**
2023 – 2025	**Promoting the Profession Sub-Committee** — MTL Programme **Field Placement Committee**
2022 – 2023	**Field Placement Committee**
2021 – 2022	**Promoting the Profession Sub-Committee** — MTL Programme
2021 – 2025	**Faculty Board Member** — Faculty of Education, University of Malta

Dissertation Supervision & Examination — 2025/26

Principal/Sole Supervisor (EDU5001):

- Jean Paul Bonnici (MTL, Full-time)
- Ylenia Vella (MTL, Part-time)
- Noela Refalo (MTL, Full-time)

Examiner (EDU5001):

- Dale Joseph Ellul (MTL, Full-time)
- Dylan Andrew Cassar (MTL, Full-time)

Chair of Board of Examiners:

- Paola Portelli (MTL, Full-time)
- Josianne Grima (B.Ed. Hons., Full-time — TET3004)

Peer Review

- Informatics in Education
- Journal of Information Technology Education: Research (JITE:R)
- Australasian Journal of Gifted Education
- Universal Access in the Information Society
- Discover Education
- Journal of Academic Ethics

Community & Outreach

- Co-founder — Computational Thinking Malta (computationalthinking.mt)
- Co-organiser — Bebras Challenge Malta (national coordinator)
- Voluntary Entrepreneurship Course — National Council of Women, Malta (Sep 2025 – Jan 2026)
- Voluntary International Teaching — Computer Science & Computing Pedagogy, Pakistan (Feb 2026 – present)

Continuing Professional Development (Selected)

2026 - *Advanced Critical Thinking Webinar* | Sage Campus, February 2026
Navigating the NEO & Neuroticism Webinar | Hogrefe, February 2026

2025 - *AI & Scientific Writing* | Springer Nature, November 2025
Human in the Loop: Equipping Peer Reviewers to Detect AI-Generated Content | IGI Global, September 2025
AI in Psychological Report Writing: Best Practices, Validation, and Ethical Use (Part 1) | PARtalks, September 2025
Navigating the Ethical Landscape: Exploring AI in School Psychology | PARtalks, April 2025

Harnessing Neurodivergence in University Students | Sage, March 2025
Assessing Emotion Regulation with the FEEL-KJ-2 | Hogrefe, May 2025
Introducing PACT: Autism-Affirming Communication Therapy for Children | Hogrefe, May 2025
AI in Contemporary Education | Identific, May 2025
AI and Ethics in Scholarly Publishing | IGI Global, September 2024

2024- *How to Do Research and Get Published Webinar Series* | Sage, March 2024
How to Write a Paper: Qualitative Methodology | Sage, March 2024

2023 – 2024 - *Foundations of University Teaching and Learning* | University of Malta, September 2023 – February 2024

2023 - *Data Science Summer School* | University of Malta, July 2023 (in-person)

2021 - *IBM SPSS: Introduction A* | University of Malta, November 2021

PREVIOUS EMPLOYMENT

Sep 2020 – Sep 2021

Learning Professional Development Expert (Casual)
Institute for Education, Malta

- Developed and delivered pedagogy modules for the MEd in Computing, ICT and Vocational and Applied IT
- All modules designed in line with Government policy and accredited by NCFHE

Oct 2019 – May 2021

IT Master & Member of SMCHS IT Leadership Team
St. Martin's College, Swatar, Malta

- Co-developed a 3-year IT transformational vision across St. Martin's College and Chiswick House School
- Led school's work towards Microsoft Showcase School status

Sep 2017 – May 2021

Full-time Learning Area Leader — Mathematics, Commerce & ICT
St. Martin's College, Swatar, Malta

- Led a department of 12 teachers across Mathematics, Accounting, Economics, Computing and IT
- Served as Independent Schools' representative on the MATSEC Board for O Level and A Level Computing

Sep 2015 – May 2021

Full-time Computing Lecturer
St. Martin's College, Swatar, Malta

Delivered the Computing curriculum at Senior School (Years 9–11) and Sixth Form (Years 12–13), covering O Level and A Level Advanced Computing.

Oct 2017 – Jan 2018

Game-Based Learning Consultant
MITA, Malta

Educational and pedagogical consultant for School Invaders (GBL4ESL Erasmus+ project). Applied Bloom's Taxonomy to model pedagogical content for O Level Computing games.

Sep 2001 – Jun 2006

Post-Secondary Teacher — A Level Computing
St. Aloysius Sixth Form College, Birkirkara, Malta

Delivered A Level Computing curriculum combining didactic teaching, small group discussions, and practical sessions. Mentored students and provided formal career guidance.

Sep 2001 – Jun 2006

Associate Lecturer in Education (Part-time)

University of Malta, Msida

Delivered coursework for trainee teachers in educational software use (4 ECTS, every semester).

EDUCATION & TRAINING

2006 – 2013

Doctor of Philosophy

Queen's University Belfast, United Kingdom

Thesis: Mainstream Technologically Enhanced Learning Environments in Special Schools in Northern Ireland. Funded by competitive PhD studentship. Awarded July 2013.

2002 – 2005

MSc in Advanced Learning Technology

Lancaster University, United Kingdom

1997 – 2001

Bachelor of Education (Honours) — Second Class Upper Division

University of Malta

Teaching Area: Computing

LANGUAGES & SKILLS

Languages: Maltese (native), English (native), Italian (intermediate), French (basic)

Digital Competence: Proficient across information processing, communication, content creation, safety, and problem solving (EU DigComp framework)